

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 124 — Past Questions & Materials
by Renance

Course: CSC 124
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 100

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Paper 1 of 1 — CSC 124

Level 100 • Second Semester • Handout

Lecture Notes on

Business Intelligence
and Analytics

FUPRE-CSC 124: Business Intelligence and Analytics

(2 unit; Core; L = 30; P = 0)

Senate-approved relevance: Business intelligence explores the impact of computing and data to businesses and immediate community. It explores the intelligent practices as frontiers to push computing

that seeks to bring about an improvement in the financial portfolios as well as reveals the social orders

and cultural practices within which computer and information helps provide intelligence to ease the business world. Students are to read, grasp and understand the critical role of computing to contemporary

businesses. Students are to explore the impact of business automation and data intelligence to capitalist

manufacturing, academics, and commercial projects. The making of new social and economic norms,

explore the worlds of techno-bodies, virtual sociality and new material substrates for computing.

Emphasis is placed on how ideas about gender and other social differences shape labor practices, models

of cognition, and material and symbolic practices of networking in our society today.

Overview: The curriculum for B.Sc. Computer Science prepares a Computer Scientist for employment in

Sectors and Industries geared towards the development/deployment of innovations and researches that

will assist in circular sustainable economy. The objectives of the course, learning outcomes, and contents

are provided to ensure that students have a well-grounded knowledge with the application of business

practices and principles to benefit the society in this revolution with industrial innovations giving way to

transformation of the digital era with the advancements of the Internet and social networks, which has

today become a way of life and culture. The course will also prepare the students to find a place in waste

management as all industries generate waste in the process of production.

Objectives

1. Discuss Business Intelligence
2. Discuss gathering of data to aid business intelligence practices

3. Demonstrate knowledge discovery in data model as active ingredient for business practices in the oil and gas sector especially for multinationals
4. Discuss the various strategies for Business intelligence and management approaches
5. Understand the nature and structure of data predictive models as used for business practices
6. Demonstrate the use of some BI system and platforms
7. Understand parameters for successful performance of data mining approaches in business practices

Learning outcomes

On completion of the course, students should be able to:

1. Discuss Business Intelligence
2. Discuss gathering of data to aid business intelligence practices
3. Demonstrate knowledge discovery in data model as active ingredient for business practices in the oil and gas sector especially for multinationals
4. Discuss the various strategies for Business intelligence and management approaches
5. Understand the nature and structure of data predictive models as used for business practices
6. Demonstrate the use of some BI system and platforms
7. Understand parameters for successful performance of of data mining approaches in business practices

Course contents

1. Business Intelligence (BI): Definition, Types, Evolution and Application
2. Data and Types of Information Processing
3. Common Data Problems
4. Knowledge Discoveries in Data (KDD) models
5. Decision Making in Digital Revolution Era
6. BI Components
7. BI – The general Process
8. BI Capabilities
9. BI systems and platforms
10. BI Architectures
11. Data Gathering, OLAP and Analysis Techniques
12. Data Mining: Processes and Methods
13. Learning modes and strategies
14. Data Reduction: Adaptive-Boost and XGBoost
15. Model Validation
16. Logistic regression
17. Market Basket Analysis
18. Predictive Data Analysis: Definition, Techniques, Algorithms and Implementations
19. Classification Models
20. Data Presentation and Visualization with Python

21. Business Implications for BI and Data Mining

22. Changing BI Platforms

1

BUSINESS INTELLIGENCE

Business intelligence (BI) combines business analytics, data mining, data visualization, data tools and infrastructure, and best practices to help organizations make more data-driven decisions. In practice, you know you've got modern business intelligence when you have a comprehensive view of your organization's data and use that data to drive change, eliminate inefficiencies, and quickly adapt to market or supply changes. Modern BI solutions prioritize flexible self-service analysis, governed data on trusted platforms, empowered business users, and speed to insight.

Business Intelligence is a set of processes, architectures, and technologies that convert raw data into meaningful information that drives profitable business actions. It is a suite of software and services to transform data into actionable intelligence and knowledge. BI has a direct impact on organization's strategic, tactical and operational business decisions. BI supports fact-based decision making using historical data rather than assumptions and gut feeling. BI tools perform data analysis and create reports, summaries, dashboards, maps, graphs, and charts to provide users with detailed intelligence about the nature of the business.

Why is BI important?

- i. Measurement: creating KPI (Key Performance Indicators) based on historic data.
- ii. Identify and set benchmarks for varied processes.
- iii. With BI systems organizations can identify market trends and spot business problems that need to be addressed.
- iv. BI helps on data visualization that enhances the data quality and thereby the quality of decision making.
- v. BI systems can be used not just by enterprises but SME (Small and Medium Enterprises)

How Business Intelligence systems are implemented?

step 1) Raw Data from corporate databases is extracted. The data could be spread across multiple systems heterogeneous systems.

step 2) The data is cleaned and transformed into the data warehouse. The table can be linked, and data cubes are formed.

Step 3) Using BI system the user can ask queries, request ad-hoc reports or conduct any other analysis.

Examples of Business Intelligence System used in Practice

Example 1: In an Online Transaction Processing (OLTP) system information that could be fed into product database could be

- i. add a product line
- ii. change a product price

Correspondingly, in a Business Intelligence system query that would be executed for the product

subject area could be did the addition of new product line or change in product price increase revenues

In an advertising database of OLTP system query that could be executed

□ Changed in advertisement options

□ Increase radio budget

Correspondingly, in BI system query that could be executed would be how many new clients added due to change in radio budget.

In OLTP system dealing with customer demographic data bases data that could be fed would be

□ increase customer credit limit

□ change in customer salary level

Correspondingly in the OLAP system query that could be executed would be can customer profile changes support higher product price.

Example 2: A hotel owner uses BI analytical applications to gather statistical information regarding average occupancy and room rate. It helps to find aggregate revenue generated per room. It also collects statistics on market share and data from customer surveys from each hotel to decides its competitive position in various markets. By analyzing these trends year by year, month by month and day by day helps management to offer discounts on room rentals.

Example 3: A bank gives branch managers access to BI applications. It helps branch manager to determine who are the most profitable customers and which customers they should work on. The use of BI tools frees information technology staff from the task of generating analytical reports for the departments. It also gives department personnel access to a richer data source.

Four types of BI users

There are four key players who use Business Intelligence System:

1. The Professional Data Analyst: The data analyst is a statistician who always needs to drill deep down into data. BI system helps the data analyst to get fresh insights to develop unique business strategies.
2. The IT users: The IT user also plays a dominant role in maintaining the BI infrastructure.
3. The head of the company: CEO can increase the profit of their business by improving operational efficiency in their business.
4. The Business Users: Business intelligence users can be found from across the organization.

There are mainly two types of business users:

- i. Casual business intelligence user
- ii. The power user.

The difference between both of them is that a power user has the capability of working with complex data sets, while the casual user need will make him use dashboards to evaluate predefined sets of data.

Advantages of Business Intelligence

Here are some of the advantages of using Business Intelligence System:

1. Boost productivity: With a BI program, It is possible for businesses to create reports with a

single click thus saves lots of time and resources. It also allows employees to be more productive on their tasks.

2. To improve visibility: BI also helps to improve the visibility of these processes and make it possible to identify any areas which need attention.

3. Fix Accountability: BI system assigns accountability in the organization as there must be someone who should own accountability and ownership for the organization's performance against its set goals.

4. It gives a bird's eye view: BI system also helps organizations as decision makers get an overall bird's eye view through typical BI features like dashboards and scorecards.

5. It streamlines business processes: BI takes out all complexity associated with business processes. It also automates analytics by offering predictive analysis, computer modeling, benchmarking and other methodologies.

6. It allows for easy analytics. BI software has democratized its usage, allowing even nontechnical or non-analysts users to collect and process data quickly. This also allows putting the power of analytics from the hands of many people.

BI System Disadvantages

1. Cost: Business intelligence can prove costly for small as well as for medium-sized enterprises. The use of such type of system may be expensive for routine business transactions.

2. Complexity: Another drawback of BI is its complexity in implementation of datawarehouse. It can be so complex that it can make business techniques rigid to deal with.

3. Limited use Like all improved technologies, BI was first established keeping in consideration the buying competence of rich firms. Therefore, BI system is yet not affordable for many small and medium size companies.

4. Time Consuming Implementation It takes almost one and half year for data warehousing system to be completely implemented. Therefore, it is a time-consuming

2

ANALYTICS

Analytics is a body of knowledge consisting of statistical, mathematical and operations research techniques, and Artificial intelligence techniques such as machine learning and deep learning algorithms, data collection and storage, data management processes such as data extraction, transformation and loading (ETL). Many companies use analytics as a competitive strategy. A typical

data-driven decision making process uses following steps:

1. Identify the problem or opportunity for value creation.
2. Identify the sources of data (primary & secondary)
3. Pre-process the data for issues such as missing and incorrect data.
4. Divide the data sets into subsets training and validation.
5. Build analytical models and identify the best model using model performance in validation data.
6. Implement solution/Decision/Develop product.

Data Science

Data Science is an interdisciplinary field. We can consider it to be an amalgamation of different fields such as data manipulation, data visualization, statistical analysis, and Machine Learning. Each of these sub-domains has equal importance.

Data Manipulation

With the help of data manipulation techniques, you can find interesting insights from the raw data with minimal effort. Data manipulation is the process of organizing information to make it readable and understandable. Engineers and scientists perform data manipulation using data manipulation language (DML) capable of adding, deleting, or altering data. Data comes from various sources. While working with disparate data, you need to organize, clean, and transform it to use it in your decision-making process. This is where data manipulation fits in. Data manipulation allows you to manage and integrate data helping drive actionable insights. Data manipulation, also known as data

preparation, enables users to turn static data into fuel for business intelligence and analytics. Many data scientists use data preparation software to organize data and generate reports, so non-analysts

and other stakeholders can derive valuable information and make informed decisions.

Importance of Data manipulation

Data manipulation makes it easier for organizations to organize and analyse data as needed. It helps

them perform vital business functions such as analyzing trends, buyer behaviour, and drawing insights from their financial data. Data manipulation offers several advantages to businesses, including:

Consistency:

Data manipulation maintains consistency across data accumulated from different sources, giving businesses a unified view that helps them make better, more informed decisions.

Usability:

Data manipulation allows users to cleanse and organize data and use it more efficiently.

Forecasting:

Data manipulation enables businesses to understand historical data and helps them prepare future forecasts, especially in financial data analysis.

Cleansing:

Data manipulation helps clear unwanted data and keep information that matters. Enterprises can clean up records, isolate, and even reduce unnecessary variables, and focus on the data they need.

Data visualization:

It is the practice of converting raw information (text, numbers, or symbols) into a graphic format. The data is visualized with a clear purpose: to show logical correlations between units, and define inclinations, tendencies, and patterns. Depending on the type of logical connection and the data itself,

visualization can be done in a suitable format. So, it's simple, any analytical report contains examples of data interpretations like pie charts, comparison bars, demographic maps, and much

more. As we've mentioned, a data representation tool is just the user interface of the whole business

intelligence system. Before it can be used for creating visuals, the data goes through a long process.

This is basically a description of how Business Analytics works, so we'll break it down into the stages:

1. First things first, you should define data sources and data types that will be used. Then transformation methods and database qualities are determined.
2. Following that, the data is sourced from its initial storages, for example, Google Analytics, ERP, CRM, or SCM system.
3. Using API channels, the data is moved to a staging area where it is transformed. Transformation assumes data cleaning, mapping, and standardizing to a unified format.
4. Further, cleaned data can be moved into a storage: a usual database or data warehouse. To make it possible for the tools to read data, the original base language of datasets can also be rewritten.

3

TYPES OF ANALYTICS:

For different stages of business analytics huge amount of data is processed at various steps. Depending on the stage of the workflow and the requirement of data analysis, there are four main kinds of analytics – descriptive, diagnostic, predictive and prescriptive. These four types together answer everything a company needs to know- from what's going on in the company to what solutions to be adopted for optimising the functions. The four types of analytics are usually implemented in stages and no one type of analytics is said to be better than the other. They are interrelated and each of these offers a different insight. With data being important to so many diverse sectors- from manufacturing to energy grids, most of the companies rely on one or all of these types of analytics. With the right choice of analytical techniques, big data can deliver richer insights for the companies Before diving deeper into each of these, let's define the four types of analytics:

- i) Descriptive Analytics: Describing or summarizing the existing data using existing business intelligence tools to better understand what is going on or what has happened.
- ii) Diagnostic Analytics: Focus on past performance to determine what happened and why. The result of the analysis is often an analytic dashboard.
- iii) Predictive Analytics: Emphasizes on predicting the possible outcome using statistical models and machine learning techniques.
- iv) Prescriptive Analytics: It is a type of predictive analytics that is used to recommend one or more course of action on analyzing the data.

Let's understand these in a bit more depth.

3.1. Descriptive Analytics

This can be termed as the simplest form of analytics. The mighty size of big data is beyond human comprehension and the first stage hence involves crunching the data into understandable

chunks. The purpose of this analytics type is just to summarise the findings and understand what is going on. Among some frequently used terms, what people call as advanced analytics or business intelligence is basically usage of descriptive statistics (arithmetic operations, mean, median, max, percentage, etc.) on existing data. It is said that 80% of business analytics mainly involves descriptions based on aggregations of past performance. It is an important step to make raw data understandable to investors, shareholders and managers. This way it gets easy to identify and address the areas of strengths and weaknesses such that it can help in strategizing. The two main techniques involved are data aggregation and data mining stating that this method is purely used for understanding the underlying behavior and not to make any estimations. By mining historical data, companies can analyze the consumer behaviors and engagements with their businesses that could be helpful in targeted marketing, service improvement, etc. The tools used in this phase are MS Excel, MATLAB, SPSS, STATA, etc

3.2 Diagnostic Analytics

Diagnostic analytics is used to determine why something happened in the past. It is characterized by techniques such as drill-down, data discovery, data mining and correlations. Diagnostic analytics takes a deeper look at data to understand the root causes of the events. It is helpful in determining what factors and events contributed to the outcome. It mostly uses probabilities, likelihoods, and the distribution of outcomes for the analysis. In a time series data of sales, diagnostic analytics would help you understand why the sales have decrease or increase for a specific year or so. However, this type of analytics has a limited ability to give actionable insights. It just provides an understanding of causal relationships and sequences while looking backward. A few techniques that uses diagnostic analytics include attribute importance, principle components analysis, sensitivity analysis, and conjoint analysis. Training algorithms for classification and regression also fall in this type of analytics.

3.3 Predictive Analytics

As mentioned above, predictive analytics is used to predict future outcomes. However, it is important to note that it cannot predict if an event will occur in the future; it merely forecasts what are the probabilities of the occurrence of the event. A predictive model builds on the preliminary descriptive analytics stage to derive the possibility of the outcomes. The analytics is found in sentiment analysis where all the opinions posted on social media are collected and analyzed (existing text data) to predict the person's sentiment on a particular subject as being-positive, negative or neutral (future prediction). Hence, predictive analytics includes building and validation of models that provide accurate predictions. Predictive analytics relies on machine learning algorithms like random forests, SVM, etc. and statistics for learning and testing the data. Usually, companies need trained data scientists and machine learning experts for building these models. The most popular tools for predictive analytics include Python, R, RapidMiner, etc. The prediction of future data relies on the existing data as it cannot be obtained otherwise. If the model is properly tuned, it can be used to support complex forecasts in sales and marketing. It goes a step ahead of the standard BI in giving accurate predictions.

3.4 Prescriptive Analytics

The basis of this analytics is predictive analytics but it goes beyond the three mentioned above to suggest the future solutions. It can suggest all favorable outcomes according to a specified course of action and also suggest various course of actions to get to a particular outcome. Hence, it uses a strong feedback system that constantly learns and updates the relationship between the action and the outcome. The computations include optimisation of some functions that are related to the desired outcome. For example, while calling for a cab online, the application uses GPS to connect

you to the correct driver from among a number of drivers found nearby. Hence, it optimizes the distance for faster arrival time. Recommendation engines also use prescriptive analytics. The other approach includes simulation where all the key performance areas are combined to design the correct solutions. It makes sure whether the key performance metrics are included in the solution. The optimisation model will further work on the impact of the previously made forecasts. Because of its power to suggest favorable solutions, prescriptive analytics is the final frontier of advanced analytics or data science, in today's term. The four techniques in analytics may make it seem as if they need to be implemented sequentially. However, in most scenarios, companies can jump directly to prescriptive analytics. As for most of the companies, they are aware of or are already implementing descriptive analytics but if one has identified the key area that needs to be optimised and worked upon, they must employ prescriptive analytics to reach the desired outcome. According to research, prescriptive analytics is still at the budding stage and not many firms have completely used its power. However, the advancements in predictive analytics will surely pave the way for its development.